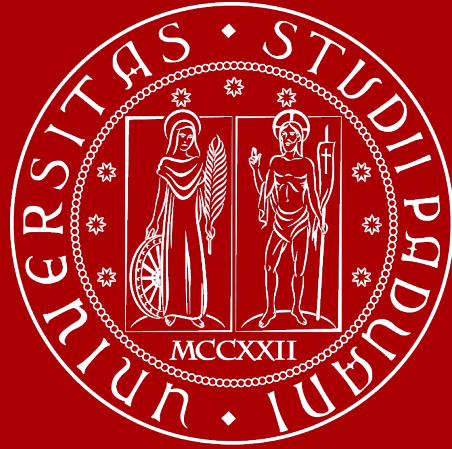


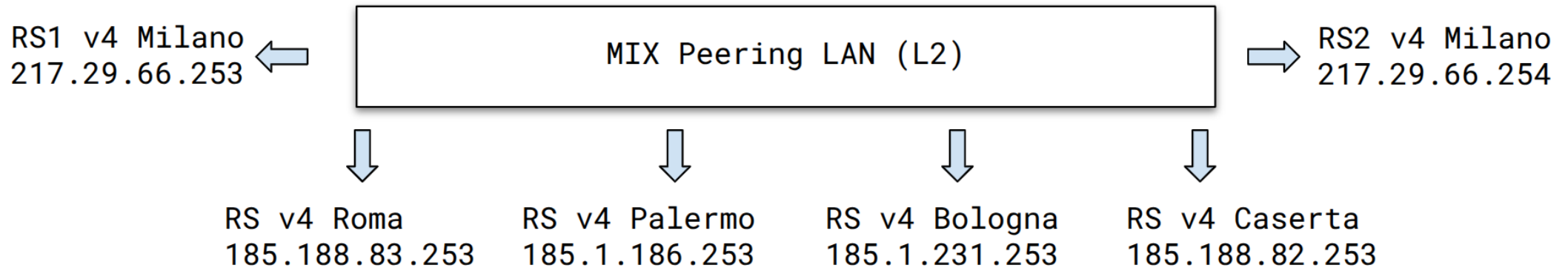


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BPG Metrics for Wide-Area Internet Exchange Points

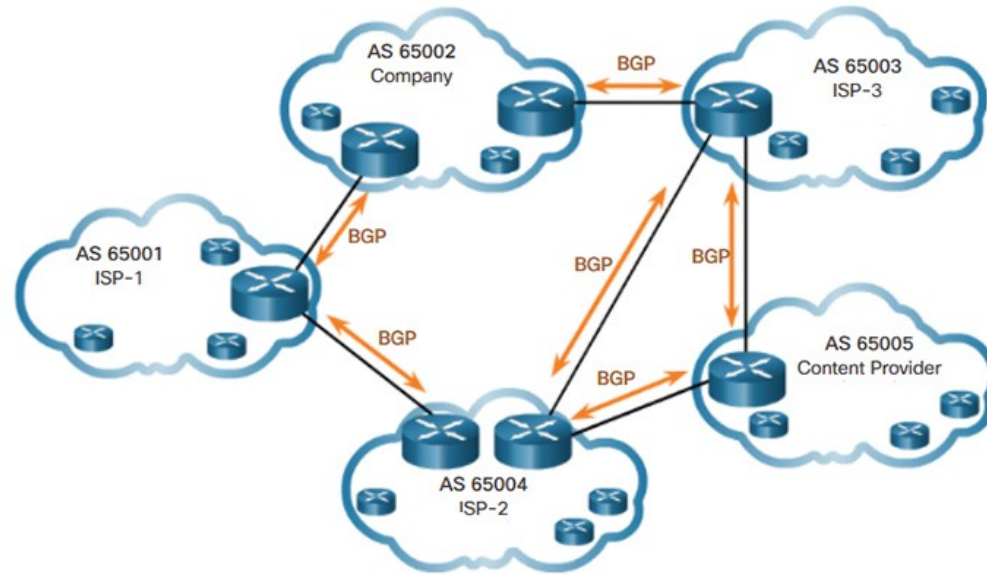
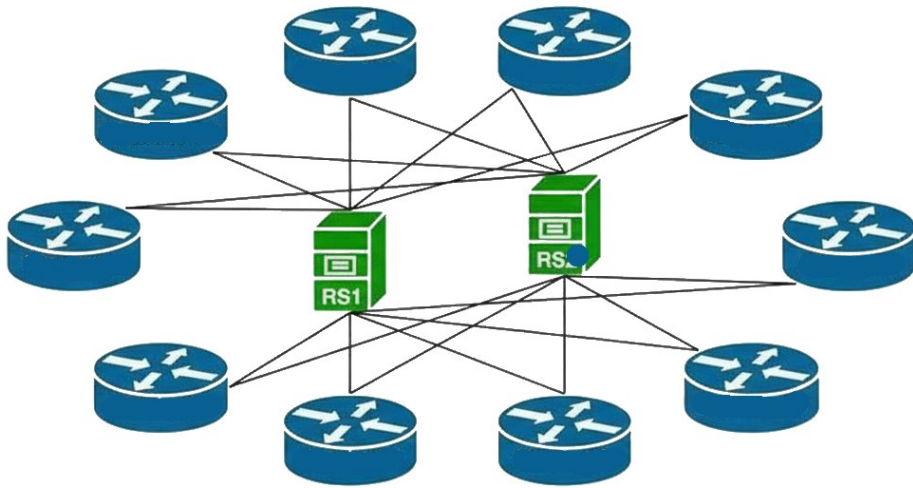


Introduction



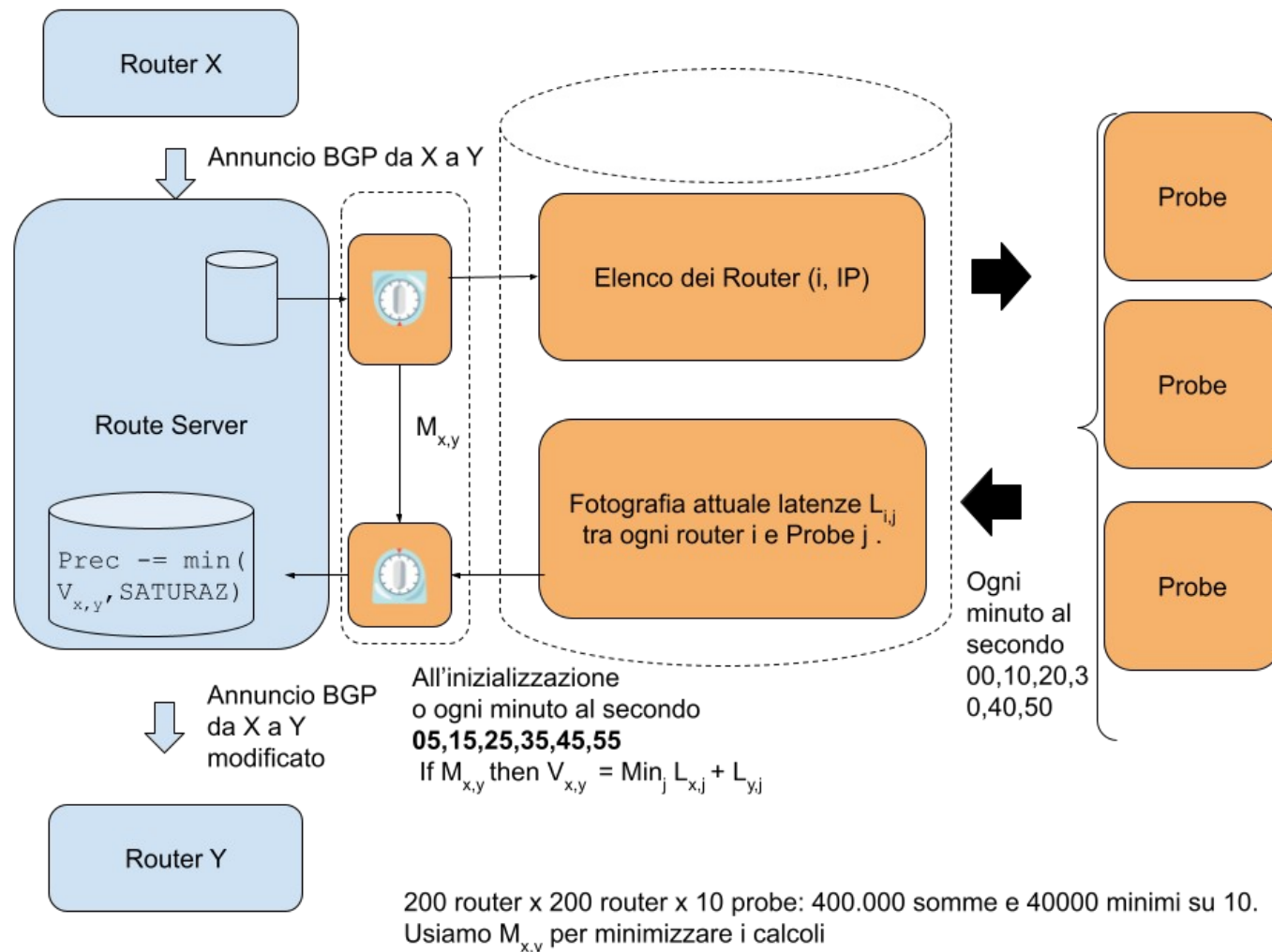
The MIX implements a Layer 2 switching infrastructure that, through Virtual LAN Emulation techniques, gives users the impression of a single shared LAN, even though it is geographically distributed across multiple locations (Milan, Rome, Bologna, Palermo, and Caserta).

Route server



Through EBGP sessions, each network participating in this type of interconnection—referred to as a “route server client”—sends the RS information regarding the reachability of its network prefixes. The route server **does not directly carry data traffic**: all routes learned from peers are stored in an internal RS database called the Loc-RIB (Local Routing Information Base).

Solution Developed



BIRD is a routing daemon: a non-interactive program that communicates with other routers and calculates routing tables.



- I retrieve data on the routers connected to the route server (IP address and protocol name) by **querying Bird's configuration** file;
- I save this data in a table named "bgp_connections," which is **synchronized** between the probe and the route server.

Tabella 6.1: Esempio di tabella **bgp_connections** con tre router distinti collegati al router server.

id	protocol_name	peer_ip	last_seen
1	router1	192.168.122.178	2025-10-20 12:50:55
2	router3	192.168.122.52	2025-10-20 12:50:55
3	router2	192.168.122.5	2025-10-20 12:50:55



- Each probe, acting completely independently, **pings every router listed in “bgp_connections”** and loads the ping results into the “ping_results” table, which is also synchronized between the probes and the route server;

Tabella 6.2: Esempio della tabella **ping_results** con due probe ed i tre router già presentati nell’output del comando **sudo birdc show protocols all** prima.

id	probe_ip	target_ip	timestamp	avg_ping	mdev_ping	packets_sent
2	192.168.122.80	192.168.122.178	2025-10-20 12:50:50	1.39	0.12	6
4	192.168.122.80	192.168.122.5	2025-10-20 12:50:50	1.10	0.07	6
6	192.168.122.80	192.168.122.52	2025-10-20 12:50:50	1.12	0.17	6
7	192.168.122.67	192.168.122.178	2025-10-20 12:50:50	1.50	0.17	6
9	192.168.122.67	192.168.122.5	2025-10-20 12:50:50	1.05	0.21	6
11	192.168.122.67	192.168.122.52	2025-10-20 12:50:50	0.99	0.13	6

- I obtain the list of exports between peer routers from Bird in order to **significantly optimize** the peer-to-peer calculation of latency for each probe;
- For each router, I select the **lowest latency** time and adjust the preference value accordingly.



BGP local preference

$$\text{pref}(Y) = \begin{cases} 100, & T \leq 1, \\ 100 - S \cdot \frac{T-1}{49}, & 1 < T < 50, \\ 100 - S, & T \geq 50. \end{cases}$$

```
filter export_192_168_122_178 {  
    # Per rotte provenienti da 192.168.122.5: 4.890 ms -> preference 96  
    if (from = 192.168.122.5) then {  
        bgp_local_pref = 96;  
    }  
    # Per rotte provenienti da 192.168.122.52: 2.510 ms -> preference 98  
    if (from = 192.168.122.52) then {  
        bgp_local_pref = 98;  
    }  
    accept;  
}
```

- S = the **maximum penalty** to be applied (increase this value to make the preference values more spread out).
- T = total **latency**.

BGP Local Preference is a standard BGP attribute used to indicate “how much to prefer this route over other BGP routes” within the AS.



Alternative to BGP large communities

```
filter export_192_168_122_178 {  
    # Latenza tra peer to peer di circa 2ms  
    if (from = 192.168.122.5) then {  
        bgp_large_community.add((65000, 1003, 2));  
    }  
    # Latenza peer to peer di circa 1ms  
    if (from = 192.168.122.52) then {  
        bgp_large_community.add((65000, 1003, 1));  
    }  
    accept;  
}
```

In 2018, Euro-IX attempted to standardize large BGP communities. Of particular interest to us is the community **RS:1003:ms**, which stands for “IXP measured round-trip time for peer in ms.”



Route flapping

Frequent routing fluctuations (route flapping) cause network instability by increasing the load on routers and degrading performance due to constant route changes. This phenomenon occurs in scenarios such as comparable **latencies between routes, unstable nodes, or periodic load fluctuations**. Although the described algorithm uses moving averages, it lacks a hysteresis mechanism to prevent fluctuations caused by minor variations. Furthermore, a dynamic latency-based `local_pref` conflicts with existing routing policies, risking the prioritization of performance over business priorities, reliability, or capacity, which could lead to contract violations and indirect costs.



Conclusions

There are many **advantages** to this implementation:

- The gains achieved by each router are **reflected across the entire path** of packets and requests;
- **Reduced operating costs** thanks to optimized hardware utilization;
- **Better load balancing** across the network infrastructure;
- An evolution of the BGP protocol that is **easily adoptable by peer routers** (they do not need to make substantial changes) and they can still choose whether or not to ignore the suggested preference.